Last Time

Kantorovich's Problem:

(Discrete version)

$$\min_{\pi \in \Pi(\mathbf{a},\mathbf{b})} \sum_{i=1}^{n} \sum_{j=1}^{m} \mathbf{C}_{i,j}\, \pi_{i,j}$$

Kantorovich's Problem:

(Continuous version)

$$\min_{\pi \in \Pi(\alpha,\beta)} \int_{X \times Y} c(x,y)\, \mathrm{d}\pi(x,y)$$

- **Feasible set:** couplings with marginals α, β
- **Interpretation:** average "work" to move mass $\mathbb{E}_\pi(c(X,Y))$
- **Existence:** guaranteed to have a solution, but not necessarily unique

Today

- From **Kantorovich to Wasserstein:** definition and special cases
- **Metric Properties**: axioms, Wasserstein space and its geometry
- **Duality**: an alternative view of the problem